Site Weather Hazard Prediction System — Requirements Specification (v1)

Oct 31, 2026 · @Someone

This specification defines what v1 of the Site Weather Hazard Prediction System must do for the business, its users and its operators: 9 business requirements, 56 functional requirements and 25 non-functional requirements.

1. Document control

Item	Value
System	Site Weather Hazard Prediction System
Version	v1 (working prototype)
Site	Karur, Tamil Nadu (inland), one kit `kit01`
Build status	Phases 0–5 complete and tested; Phase 6 (real kit) in progress
Audience	Project guide and evaluators, developers who build or extend the system
Source documents	`docs/analysis/analysis_v1.md` (why), `docs/requirements_v1.md` (what), `docs/design/system_design_v1.md` (how), `docs/mqtt_contract.md` (kit messages)

Priority key. Must = required for v1. **Should** = wanted; can slip if time is short. Requirement IDs (for example `FR-ING-03`) are used to trace each requirement to code and tests.

2. Project overview

2.1 Problem statement

Campuses and tech parks face local heat and strong-wind hazards that affect outdoor work, events, rooftop equipment and buildings. Public IMD forecasts cover whole districts, are issued a few times a day and are not tied to conditions on the site itself.

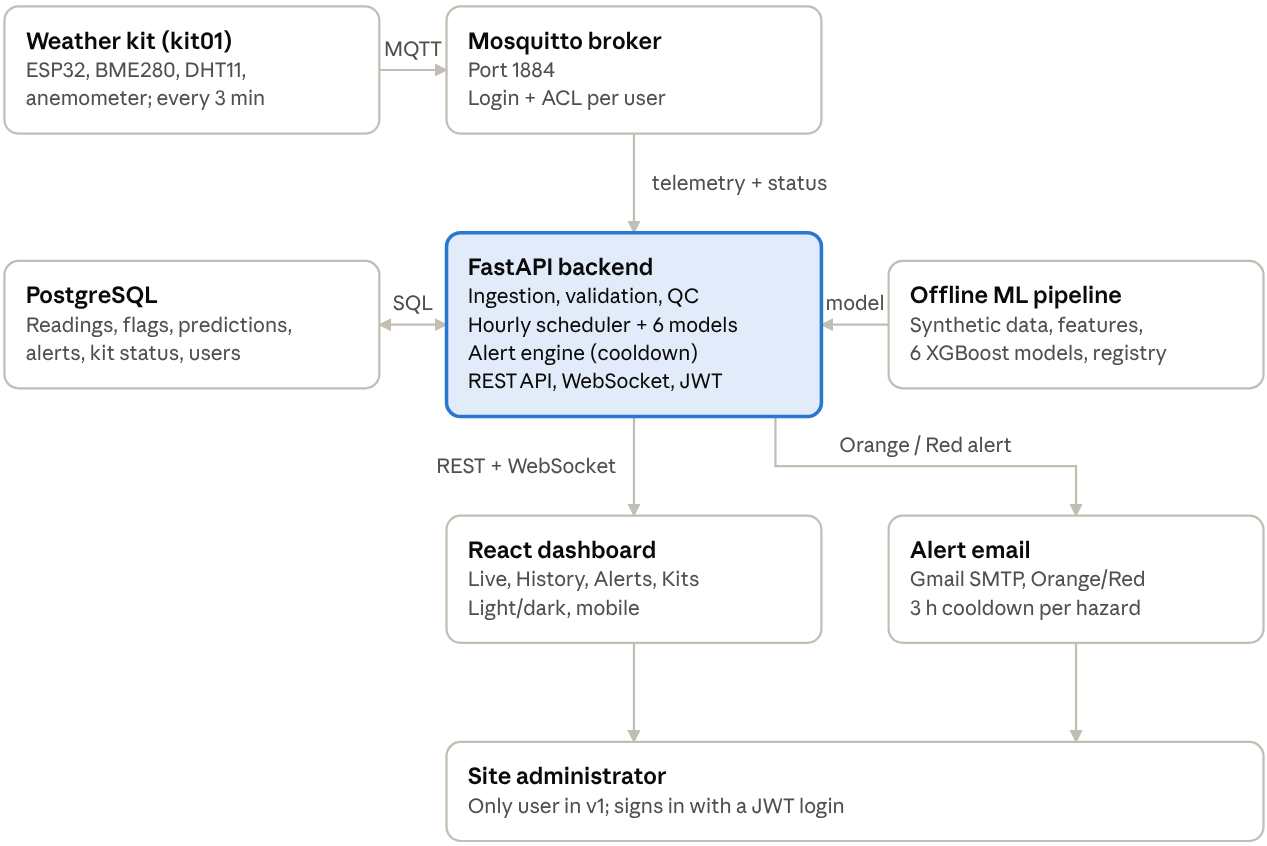
Current option	Limitation
IMD district forecasts	District-level, not site-specific, issued a few times a day
Weather apps	Model grids tens of km wide; no local sensor; no site alerts
Manual observation	No early warning; depends on someone noticing
Stand-alone weather station	Shows current values only; no prediction or alerts

Gap: no low-cost system combines a site's own live measurements with automatic short-term hazard prediction and alerts.

2.2 Solution summary

An on-site kit (ESP32 with BME280, DHT11 and an anemometer) sends readings over MQTT. A FastAPI backend checks and stores them in PostgreSQL. Every hour, six XGBoost models predict heat and high-wind risk for the next 3 hours. Results appear on an admin-only React dashboard, and Orange or Red risk is emailed to the admin.

Kit readings flow through one backend to the dashboard and email



system context · kit to admin

The backend is the only part that touches the database; the kit never talks to the dashboard directly, and models are trained offline and loaded by version.

3. Business requirements

The business need is early, site-specific warning of heat and high wind, at low cost, built so that real site data can train better models later.

3.1 Business objectives

1. Give the site administrator a live view of weather at the exact site location.
2. Warn the administrator of heat or high-wind risk up to 3 hours ahead, so outdoor work or events can be moved and rooftop items secured.
3. Deliver serious warnings (Orange, Red) by email, so the administrator does not need to watch the dashboard.

4. Collect clean, labelled real kit data to train the next version's models.
5. Keep running costs near zero: free software, a low-cost kit and an existing Windows PC.

3.2 Stakeholders

Stakeholder	Interest	v1 role
Site administrator	Knows conditions and risks; acts on alerts	Only user (dashboard and email)
Campus occupants (staff, students)	Safety	Indirect: admin passes on warnings
Facilities and maintenance team	Protect rooftop equipment; plan outdoor work	Future users
Project team	Build, evaluate and extend the system	Developers
Project guide and evaluators	Assess the work	Reviewers

3.3 Business requirements

ID	Requirement	Priority
BR-01	Measure temperature, humidity, pressure and wind speed at the site continuously, every 1–5 minutes	Must
BR-02	Predict heat and high-wind risk for the next 3 hours, updated every hour	Must
BR-03	Express risk as four levels (Green, Yellow, Orange, Red) based on IMD heat-wave criteria and local wind thresholds	Must
BR-04	Email the administrator automatically when risk reaches Orange or Red, without flooding the inbox	Must
BR-05	Restrict all data and controls to the administrator	Must

ID	Requirement	Priority
BR-06	Keep every reading, prediction and alert permanently for audit and future training	Must
BR-07	Let a new site be configured by editing settings files, with no code change	Must
BR-08	Run on one Windows PC with free, open-source software	Must
BR-09	Allow more kits, users and hazards (heavy rain) to be added later without redesign	Should

3.4 Business benefits

- Site-level warnings instead of district-level forecasts.
- Time to act: a 3-hour lead to move events indoors, pause rooftop work or secure equipment.
- A growing dataset from the site's own sensor, which public sources cannot provide.

3.5 Success measures for v1

Measure	Target
End-to-end flow	Real kit readings reach the dashboard, predictions run hourly, alerts are emailed
Reading latency	On the dashboard within 5 s of being sent
Prediction run time	Under 60 s per hourly run
Data kept	100% of readings, flags, predictions and alerts retained
Alert noise	No repeat email for the same or lower level of a hazard within 3 h

Prediction accuracy is **not** a v1 success measure: v1 models are trained on synthetic data and may be inaccurate on real weather.

4. Scope

v1 covers one site, one kit, two hazards and one admin, running locally on Windows.

4.1 In scope

- One site (Karur) and one kit (`kit01`).
- Hazards: heat and high wind.
- 3-hour prediction horizon, run every hour.
- Models trained on a synthetic dataset in the kit's exact message format.
- Email alerts to the single admin.
- Admin-only web dashboard with light and dark themes.
- Local deployment on Windows 11, no Docker.

4.2 Out of scope

Item	Planned for
Heavy-rainfall prediction and rain gauge (DFRobot SEN0575)	Later version
Retraining on real kit data	Next version
Multiple kits per site (data model already keyed by `kit_id`)	Next version
Multiple users and roles	Later version
Editing thresholds from the dashboard	Later version
Minute-level ML features	Later version
Other model types, hyper-parameter tuning, probability calibration	Not in v1 (XGBoost, default settings)
External forecasts (NWP), ERA5, airport data	Not planned
Docker deployment	Not in v1
Firmware changes (true gust measurement, NTP clock)	Not in v1: firmware is fixed

4.3 Assumptions

- The kit and the server PC are on the same local network.
- The server PC stays on and connected while the system is in use.
- Internet access is available for sending alert email through Gmail SMTP.
- IMD 1991–2020 normals for Tiruchirappalli represent Karur's climate closely enough for heat departures.

4.4 Constraints

Constraint	Effect
Kit firmware is fixed	Wind is a 1-minute average sent every 3 minutes; no timestamp (`ts`); no offline buffering, so outages show as gaps
No historical kit data	v1 models learn from synthetic data, not real site weather
Single Windows PC	No high availability; the system is down when the PC is off
Free tools only	Python, FastAPI, PostgreSQL, Mosquitto, React, XGBoost

4.5 Dependencies

Dependency	Version
Python	3.13 (3.11+ supported)
Node.js	20+
PostgreSQL	16 or 17
Mosquitto	2.x, own instance on port 1884
Gmail account	App password for SMTP

5. Functional requirements

The system has 56 functional requirements in 11 groups; 53 are Must and 3 are Should.

5.1 Configuration (FR-CFG)

ID	Requirement	Priority
FR-CFG-01	`config/site.yaml` holds all site settings (location, kits, hazard thresholds, QC limits, prediction, data sufficiency, kit status, alerts, IMD normals), so a new site needs no code change	Must
FR-CFG-02	`config/synthetic.yaml` sets synthetic data: 3 years, 5-min interval, seed, hazard rate 3–5% of hours, fault injection	Must
FR-CFG-03	Secrets (database URL, admin login and email, JWT secret, SMTP) live only in `.env`; `.env.example` has placeholders	Must
FR-CFG-04	The backend validates config at start-up and refuses to start, with a clear error, if it is invalid	Must

5.2 Kit messages (FR-MQTT)

ID	Requirement	Priority
FR-MQTT-01	Topics `weather/kit/{kit_id}/telemetry` (QoS 1) and `weather/kit/{kit_id}/status` (retained, `offline` as Last-Will); `kit_id` comes from the topic	Must
FR-MQTT-02	Payload is JSON: `temp_bme_c`, `hum_bme_pct`, `pressure_hpa`, `temp_dht_c`, `hum_dht_pct`, `wind_speed_ms`; optional `ts`, `wind_gust_ms`, `rain_mm`, `fw_version`. A failed sensor sends `null`	Must
FR-MQTT-03	Any sending interval from 1 to 5 minutes is accepted without config changes	Must

5.3 Ingestion and quality control (FR-ING, FR-QC)

ID	Requirement	Priority
FR-ING-01	An async MQTT subscriber runs inside the backend, reconnects automatically and logs connection state	Must
FR-ING-02	Each message is validated; malformed JSON or wrong types are rejected and logged; unknown kits are ignored	Must
FR-ING-03	`ts` more than 10 min in the future is rejected; missing `ts` uses arrival time and is flagged `ts_missing`; duplicates are ignored	Must
FR-ING-04	Kit status follows status messages; no reading for 10 min marks the kit offline; changes are stored and pushed live	Must
FR-QC-01	Values outside 0–55 °C, 0–100 %, 950–1050 hPa or 0–60 m/s are flagged `out_of_range`	Must
FR-QC-02	Jumps above set limits (for example > 5 °C or > 3 hPa in 5 min) are flagged `spike`	Must
FR-QC-03	The same value for 60 min or more is flagged `stuck` (calm wind of 0 excluded)	Must
FR-QC-04	BME280 is primary; DHT11 is used when BME280 is missing or flagged (`fallback`); a difference > 3 °C or > 10 % RH is flagged `bme_dht_mismatch`	Must
FR-QC-05	Flagged readings are stored with their flags, never deleted	Must

5.4 Data storage (FR-DB)

ID	Requirement	Priority
FR-DB-01	Tables: `kits`, `kit_status_log`, `readings` (raw, fused, flags), `predictions`, `alerts`, `users`	Must

ID	Requirement	Priority
FR-DB-02	`readings` is indexed on (`kit_id`, `ts`); hourly aggregates are computed by query	Must
FR-DB-03	All readings, flags, predictions and alerts are kept permanently	Must

5.5 Synthetic data and simulator (FR-SYN, FR-SIM)

ID	Requirement	Priority
FR-SYN-01	Generated rows use exactly the kit's payload fields, so one code path handles synthetic and real data	Must
FR-SYN-02	3 years at 5-min intervals (about 315,000 rows), saved to `ml/data/synthetic/`	Must
FR-SYN-03	Climate follows IMD normals for Karur: daily cycle, seasons, pressure tide, multi-day spells	Must
FR-SYN-04	Hazard events: April–May heat spells, June–September wind, pre-monsoon storms, Oct–Dec depressions; hazard hours about 3–5%	Must
FR-SYN-05	Sensor noise, DHT11 resolution and error, and injected faults (spikes, stuck values, gaps, offline periods)	Must
FR-SYN-06	The same seed and config give the same dataset	Must
FR-SIM-01	`tools/kit_simulator.py` publishes live readings in the kit's format, with options for interval, speed-up, forced hazards and offline	Must

5.6 Machine learning (FR-ML)

ID	Requirement	Priority
FR-ML-01	Labels for hour *t* come from readings in (*t*, *t* + 3 h], cumulative by level. Wind: peak ≥ 30 / 40 / 60 km/h. Heat: Yellow (Tmax ≥ 40 °C and departure ≥ 4.5 °C) or ≥ 45 °C; Orange (departure ≥ 6.5 °C) or ≥ 45 °C; Red ≥ 45 °C	Must
FR-ML-02	Hourly features from kit data only: current values, dew point, departure, lags (1–24 h), 1 h and 3 h tendencies, 6/12/24 h rolling stats, time encodings; the same code for training and live use	Must
FR-ML-03	Six binary XGBoost classifiers (3 levels × 2 hazards), default settings, random 80/20 split	Must
FR-ML-04	Each model records precision, recall, F1, PR-AUC, Brier score and top features, with a note that scores are synthetic	Should
FR-ML-05	Each training run saves to `ml/models/v{N}/` with metadata; an `active` pointer picks the live version; every prediction records it	Must
FR-ML-06	One command each to generate, build, train and evaluate	Must

5.7 Inference (FR-INF)

ID	Requirement	Priority
FR-INF-01	Predictions run every hour for each enabled kit; a manual "run now" exists for testing	Must
FR-INF-02	Gaps ≤ 30 min are interpolated. More than 25% of the last 24 h missing, or no data in the last 60 min → status `insufficient_data`, no alert	Must
FR-INF-03	Probabilities are made monotonic per hazard; level = highest with probability ≥ 0.50, else Green; valid for 3 h; `fallback_used` is recorded	Must

ID	Requirement	Priority
FR-INF-04	Predictions are stored, pushed live and passed to the alert engine	Must

5.8 Alerts (FR-ALT)

ID	Requirement	Priority
FR-ALT-01	An alert is raised when a hazard reaches Orange or Red (configurable)	Must
FR-ALT-02	Email goes to the admin address in `.env`	Must
FR-ALT-03	After an email, the same or a lower level of that hazard is not emailed for 3 h (recorded as `cooldown`); a higher level is emailed at once	Must
FR-ALT-04	Subject `[<Level>] <Hazard> risk at <Site> — next 3 h`; body has level, probabilities, valid-until (IST), latest readings, model version, notes and a dashboard link	Must
FR-ALT-05	SMTP failures are logged and recorded as `failed`, without stopping the pipeline	Must
FR-ALT-06	`alerts.enabled` switches email on or off; when off, alerts are still recorded as `disabled`	Must

5.9 Authentication (FR-AUTH)

ID	Requirement	Priority
FR-AUTH-01	One admin account, created on first start from `.env`; password stored as a bcrypt hash	Must
FR-AUTH-02	Login returns a JWT (8 h expiry); every endpoint except login and health requires it, including the WebSocket	Must
FR-AUTH-03	The `users` table and role field allow more users later without schema redesign	Should

5.10 API (FR-API)

ID	Requirement	Priority
FR-API-01	REST endpoints for login, health, kits and status history, readings (latest, range, CSV export), predictions (latest, history, run now), alerts (filtered, paginated), data quality and system info	Must
FR-API-02	WebSocket `/ws/live` pushes `reading`, `kit_status`, `prediction` and `alert` messages	Must
FR-API-03	OpenAPI docs at `/docs`	Should

5.11 Dashboard (FR-UI)

ID	Requirement	Priority
FR-UI-00	Look and layout follow `docs/design/ui_reference_v1.html`; behaviour and data follow this specification	Must
FR-UI-01	React + Vite + Recharts; light and dark modes (system default, remembered); responsive; times in IST, wind in km/h	Must

ID	Requirement	Priority
FR-UI-02	Top bar: site name, navigation, kit status and last seen, IST clock, theme toggle, logout	Must
FR-UI-03	Login page with show/hide password, error message, redirect on success, auto-logout on token expiry	Must
FR-UI-04	Live dashboard: wind and heat risk cards, live sensor cards with QC warnings, 24 h charts with BME280 and DHT11 lines and wind band lines	Must
FR-UI-05	History: preset and custom date ranges, charts, predicted-risk timeline, CSV download	Must
FR-UI-06	Alerts: table with email status, filters by hazard, level and date, pagination	Must
FR-UI-07	Kits and system: kit details, 7-day online strip, data-quality summary, system health, read-only settings, "Run prediction now"	Must

6. Non-functional requirements

The system has 25 non-functional requirements; all are Must except NFR-REL-04 and NFR-MNT-04.

6.1 Performance

ID	Requirement	Target
NFR-PER-01	A reading appears on the dashboard after it is published	Within 5 s
NFR-PER-02	An hourly prediction run for all enabled kits completes	Within 60 s
NFR-PER-03	Training all six models from the synthetic dataset (`ml.cli all`)	About 30 s on a laptop
NFR-PER-04	Storage for one kit without special tuning	About 105,000 rows a year at 5 min (175,000 at 3 min)

6.2 Reliability and availability

ID	Requirement
NFR-REL-01	The backend survives broker or database restarts and reconnects automatically
NFR-REL-02	A failed email, bad message or bad reading never stops ingestion, prediction or alerting
NFR-REL-03	Missing data produces `insufficient_data`, never a wrong prediction or alert
NFR-REL-04 (Should)	Mosquitto and PostgreSQL run as Windows services; the backend starts at login

6.3 Security

ID	Requirement
NFR-SEC-01	No secrets in source code or committed config; `.env` is git-ignored
NFR-SEC-02	Passwords stored only as bcrypt hashes

ID	Requirement
NFR-SEC-03	All API and WebSocket access needs a valid JWT, except login and health
NFR-SEC-04	The MQTT broker requires a username and password, with an ACL limiting each user's topics
NFR-SEC-05	The backend refuses to start while `.env` holds placeholder values; `JWT_SECRET` has at least 32 characters

6.4 Usability

ID	Requirement
NFR-USE-01	Light and dark themes; hazard colours readable in both
NFR-USE-02	Works on desktop and mobile; the menu collapses on small screens
NFR-USE-03	Times in IST and wind in km/h everywhere the admin sees them
NFR-USE-04	Error messages name the file and setting to fix

6.5 Maintainability and portability

ID	Requirement
NFR-MNT-01	Unit tests cover QC rules, labels, features, level selection and alert cooldown; `pytest` passes before each checkpoint
NFR-MNT-02	Logs go to the console and a rotating file (ingestion errors, QC, runs, alerts)
NFR-MNT-03	`README.md` gives step-by-step Windows setup, running and troubleshooting

ID	Requirement
NFR-MNT-04 (Should)	Code style checked with `ruff` (Python) and Prettier (frontend)
NFR-POR-01	Runs locally on Windows 11 with no Docker; PowerShell scripts set up and start each part

6.6 Data

ID	Requirement
NFR-DAT-01	All data stored in UTC and shown in IST (Asia/Kolkata)
NFR-DAT-02	Nothing is deleted automatically; raw values and flags are kept for next-version training
NFR-DAT-03	Every prediction records the model version that made it

7. Acceptance, decisions and glossary

7.1 Acceptance criteria

v1 is accepted when all nine criteria pass in the full acceptance run (plan task 6.9, scheduled after the real kit is connected).

#	Criterion	Traces to
1	Kit readings are validated, QC-flagged and stored	BR-01, FR-ING, FR-QC
2	The dashboard shows live readings and kit status without a page reload	FR-API-02, FR-UI-04
3	Every hour, wind and heat predictions are produced, stored and shown with level and probabilities	BR-02, FR-INF
4	Insufficient data gives "insufficient data", not a prediction or alert	FR-INF-02

#	Criterion	Traces to
5	A forced Orange/Red hazard emails the admin; repeats within 3 h are suppressed; escalation emails at once	BR-04, FR-ALT
6	Only the logged-in admin can reach the dashboard and API	BR-05, FR-AUTH
7	History charts any date range and exports CSV	FR-UI-05
8	Light/dark mode and mobile layout work on every page	FR-UI-01
9	All readings, flags, predictions and alerts are retained	BR-06, FR-DB-03

7.2 Decisions log

ID	Decision	Outcome
D1	Database	PostgreSQL, no TimescaleDB
D2	Email provider	Gmail SMTP with app password; Mailpit as optional test inbox
D4	Kit firmware facts	Wind is a 1-min average; sends every 3 min; no `ts`; no offline buffering (2026-10-31)
D5	Extra derived values (dew point, “feels like”) on the dashboard	Not in v1 unless requested
D6	Alerts table content	Show all alerts with status
D7	UI design reference	`docs/design/ui_reference_v1.html`
D8	IMD thresholds and Karur normals	Heat-wave departures and 40 °C floor confirmed; IMD 1991–2020 Tiruchirappalli normals
D9	Red heat threshold	Open: v1 uses ≥ 45 °C (IMD heat wave); IMD severe heat wave is ≥ 47 °C

7.3 Glossary

Term	Meaning
Kit	ESP32 board with BME280, DHT11 and an anemometer
Reading	One telemetry message from the kit
QC	Quality control: automatic checks that flag bad readings
Fallback	Using DHT11 values when the BME280 value is unusable
Normal daily max	Expected maximum temperature for a date, from IMD 30-year monthly normals
Departure	Temperature minus the normal daily max for that date
Peak wind	Highest reported wind speed in a window, used in place of a true gust
Level	Risk colour: Green, Yellow, Orange, Red
MQTT	Lightweight publish/subscribe messaging used between kit and server
JWT	Signed login token sent with each API request
IST / UTC	Data is stored in UTC and shown in IST (Asia/Kolkata)